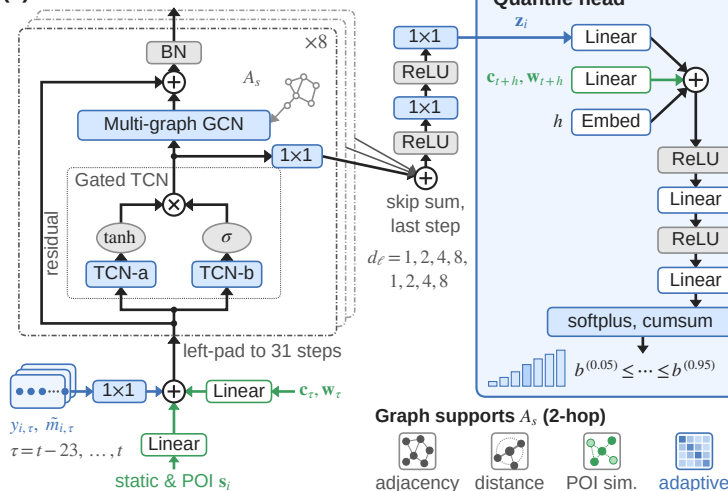


(a) Price-blind ST backbone



(b) Anchored price response

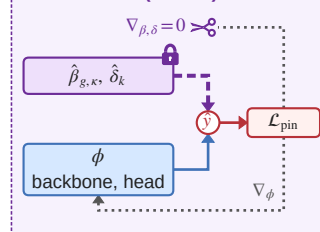
$\kappa(t+h)$	Workday		Rest day		
$g(i)$	day	night	day	night	
Low-density	○	○	○	○	
Residential	○	●	●	●	
Commercial	○	●	○	●	

(illustrative)

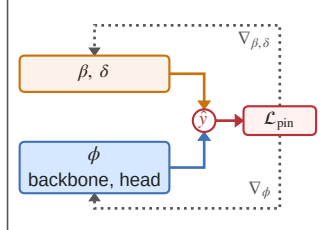
Legend:
 ● cell
 ○ context
 ○ city

$$\eta = \hat{\beta}_{g(i), \kappa(t+h)} \Delta \ell + \sum_k \hat{\delta}_k S^{(k)}$$

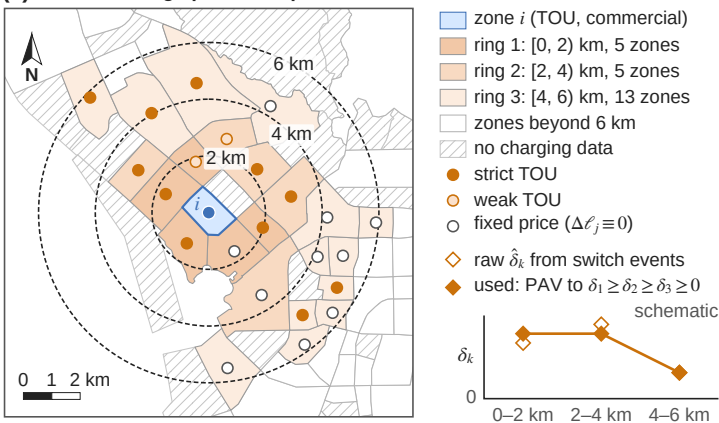
Cut feedback (default)



Joint training (ablations)



(c) Distance-ring spillover exposure



$$S_{i,t}^{(k)} = \text{mean}\{\Delta \ell_{i,t} : j \in \mathcal{R}(i,t)\}, \quad \mathcal{R}(i,t) = \{j \neq i : r_{k-1} \leq d_{ij} < r_k\}, \quad d_{ij}: \text{centroid distance}$$

$$\eta_{i,t}^{\text{spill}} = \sum_k \delta_k S_{i,t}^{(k)}, \quad \text{an empty ring gives } S_{i,t}^{(k)} = 0$$

(d) Masked adaptive conformal calibration

